

Development of Touchscreen Display Integration for Toro Diag Using Tiva Microcontrollers

Current Problem

Several current Toro/BOSS systems use Texas Instruments Tiva microcontrollers, including:

- Tiva TM4C1231H6PM
- Tiva TM4C123GH6PM
- Tiva TM4C123BH6ZRB

The TM4C1231H6PM is currently used in the Spreader Display. However, this microcontroller does not include an integrated LCD controller or hardware graphics acceleration. Because of this, an external graphics driver or display controller is required to handle display output.

This adds complexity to the design by increasing component count, cost, software development effort, and long-term maintenance requirements. It also limits future display upgrades, especially if higher-resolution or full-color touchscreen displays are desired.

A major limitation is memory. Higher-resolution displays require significantly more SRAM for frame buffering. For example, an 800 × 480 display requires approximately 768 KB for a 16-bit color frame buffer, or approximately 1.15 MB for a 24-bit color frame buffer. These requirements exceed the internal SRAM available on the TM4C123 series and may also be a limitation for other Tiva-based solutions without external memory.

Objective

The goal of this project is to simplify touchscreen display integration while maintaining compatibility with the Toro Diag diagnostic system.

The main objectives are to:

- Develop or identify a touchscreen display solution that can communicate with Toro Diag.
- Evaluate microcontroller options that include integrated display support.
- Support capacitive or resistive touchscreen input.
- Maintain CAN communication for compatibility with existing Toro/BOSS systems.
- Stay within the existing Tiva/TI ecosystem if possible.
- Identify a manageable migration path if a different microcontroller platform is required.

As part of this effort, we are currently working with Texas Instruments to identify suitable microcontroller and display options that meet the application requirements.

Key Technical Constraints

Several technical constraints must be considered when selecting a touchscreen display and microcontroller platform:

- Display resolution and color depth directly affect memory requirements.
- Higher-resolution displays may require an external graphics controller, external SRAM, or a microcontroller with an integrated LCD controller.
- CAN communication must remain available for communication with existing Toro/BOSS systems and Toro Diag.
- The selected solution must support long-term serviceability, software updates, and diagnostic compatibility.
- Touchscreen interface selection, such as capacitive versus resistive, will affect cost, durability, environmental performance, and software complexity.
- Component cost, availability, and lifecycle must be considered before selecting a final design.

What TI Provides for Tiva

Texas Instruments provides the TivaWare Graphics Library for Tiva C Series microcontrollers. This is a software-based graphics framework that supports basic graphical user interface development.

TivaWare Graphics Library includes:

- Graphics primitives such as lines, circles, text, and bitmaps
- Touch input handling for resistive touchscreen applications
- Software drivers for simple LCDs, including SPI-based and parallel display modules
- A display abstraction layer that allows graphics code to be written in a more hardware-independent way
- Example projects, documentation, and display drivers within the TivaWare for C Series package

Common examples include support for displays such as the Kentec 320 × 240 QVGA TFT display used with Tiva development hardware.

Limitations of TivaWare and TM4C123

Although TivaWare provides useful graphics support, there are several limitations when using the TM4C123 series for touchscreen display applications:

- The TM4C123 series does not include a dedicated LCD controller.
- Graphics are handled in software, which increases CPU load.
- There is no hardware graphics acceleration.
- Internal SRAM is limited, making full-frame buffering difficult or impossible for higher-resolution displays.
- The TM4C123 does not directly support full-frame RGB displays without an external display controller.
- Display resolution and frame rate are limited by MCU speed, available SRAM, and display interface bandwidth.
- Higher-resolution displays require either an onboard display controller or additional external memory and hardware.

Because of these limitations, the TM4C123 series is better suited for smaller displays or display modules that include their own graphics controller.

Options Considered

Option 1: TM4C1294NCPDT with External Memory

The TM4C1294NCPDT is a possible upgrade path within the Tiva family. This microcontroller includes an integrated LCD controller, making it more suitable for direct display control than the TM4C123 series.

This option may allow compatibility with displays such as the Newhaven NHD-7.0CTP-CAPE-P touchscreen display, depending on display interface requirements, timing, and memory needs.

However, the TM4C1294NCPDT includes only 256 KB of internal SRAM. For a larger display, such as 800 × 480, a full-frame buffer may require approximately 768 KB to 1.15 MB depending on color depth. This means external SRAM may still be required.

Advantages:

- Stays within the TI/Tiva ecosystem
- Uses familiar development tools
- Includes an integrated LCD controller
- May reduce the need for a separate graphics driver

Disadvantages:

- Internal SRAM may still be insufficient for high-resolution graphics
- External memory may be required
- Hardware design complexity may still increase

- Toro Diag compatibility would need to be verified
- May require coordination with Toro due to differences from currently deployed Tiva chips

Option 2: Grayhill Display Solution

Another option is to use a Grayhill display solution. Grayhill displays may include built-in graphics handling, touchscreen support, and CAN communication. This could reduce the amount of low-level display development required.

However, this option may require moving away from the current Toro Diag workflow. While it could simplify display development, it would likely create new integration challenges with existing diagnostic tools, software, and support processes.

Advantages:

- More complete off-the-shelf display solution
- Built-in display and touchscreen support
- May reduce low-level graphics development
- CAN-capable options are available

Disadvantages:

- May require moving away from Toro Diag
- Could require major changes to existing software and diagnostic workflows
- Higher display cost
- Increased dependency on a display vendor's software ecosystem
- Not currently considered feasible due to integration and support constraints

Option 3: Continue Using TM4C1231H6PM with External Graphics Driver

A fallback option is to continue using the TM4C1231H6PM with an external graphics driver or display controller, similar to the current approach.

This option maintains compatibility with existing hardware and software, but it does not solve the long-term limitations related to graphics performance, memory, and display flexibility.

Advantages:

- Lowest disruption to existing hardware and software
- Maintains compatibility with current Tiva-based systems
- Keeps Toro Diag integration simpler
- Builds on known architecture

Disadvantages:

- Does not solve existing memory limitations
- Requires an external graphics driver or display controller
- Limited graphics performance
- Limited flexibility for higher-resolution displays
- May increase cost and complexity as display requirements increase

Option Comparison Summary

The TM4C1294NCPDT provides the most direct path within the Tiva ecosystem because it includes an integrated LCD controller and uses familiar TI development tools. However, its internal SRAM may still be insufficient for larger displays, so external memory or graphics optimization may be required.

The Grayhill display option provides a more complete off-the-shelf display solution with built-in graphics and CAN capability. This could reduce display development effort, but it would likely require changes to existing Toro Diag workflows and long-term support processes.

Continuing with the TM4C1231H6PM and an external graphics driver provides the least disruption to current systems. However, this approach does not address the long-term issues of memory limitations, graphics performance, and display scalability.

Preliminary Recommendation

The recommended next step is to prototype with the TM4C129 development platform and verify communication with Toro Diag before committing to a full hardware redesign.

This approach keeps the project within the existing TI/Tiva ecosystem while allowing evaluation of the integrated LCD controller. The prototype should be used to determine whether the TM4C129 platform can support the required display resolution, touchscreen interface, CAN communication, and Toro Diag compatibility.

If the TM4C129 platform cannot meet performance, memory, or diagnostic compatibility requirements, then the next alternatives should be evaluated. These alternatives may include a display module with an onboard graphics controller or a higher-performance microcontroller platform such as STM32F7/H7, NXP i.MX RT, TI Sitara, or TI AM2x.

Next Steps

1. Evaluate Display Requirements

The first step is to clearly define the display requirements, including:

- Desired screen size
- Desired resolution
- Color depth
- Touchscreen type
- Environmental requirements
- Brightness and outdoor visibility needs
- Display update rate
- Required graphics complexity

2. Estimate Memory and Flash Requirements

The next step is to estimate the memory required for the selected display and graphics system.

This should include:

- SRAM required for frame buffering
- SRAM required for application code
- Flash required for fonts, icons, graphics, and firmware
- External memory requirements, if needed
- Impact of color depth on memory usage

3. Continue Microcontroller Research

Additional microcontroller options should be reviewed to determine whether they provide a better long-term solution.

Possible platforms include:

- TI TM4C129 series
- TI Hercules
- TI AM2x
- TI Sitara
- STM32F7/H7
- NXP i.MX RT series

The preferred solution should include CAN communication, display controller support, touchscreen capability, long-term availability, and manageable software development requirements.

4. Select Prototype Hardware

Prototype hardware should be selected based on display support, CAN capability, and software development resources.

The selected development platform should:

- Support the target display interface
- Provide CAN communication
- Allow touchscreen testing
- Provide enough memory for graphics testing
- Support rapid firmware development
- Allow Toro Diag compatibility testing

5. Test Toro Diag Compatibility

Compatibility with Toro Diag must be tested before selecting a final solution.

Testing should confirm:

- CAN communication works correctly
- Toro Diag can communicate with the controller
- Diagnostic messages are received and transmitted correctly
- Touch input does not interfere with diagnostic functions
- Displayed data matches expected system values
- Software updates and service workflows remain manageable

6. Review Long-Term Considerations

Before final selection, the following long-term factors should be reviewed:

- Component cost
- Component availability
- Product lifecycle
- Vendor support
- Software licensing
- Graphics library support
- Ease of future updates
- Manufacturing complexity
- Serviceability

Updates as of 5/9/2025

I spoke with Munan from Texas Instruments, and he offered to provide support in selecting suitable microcontrollers and displays for this application.

Key takeaways from the discussion:

- The TM4C129 series includes an integrated LCD controller, which may make it compatible with the originally selected display depending on the display interface, timing requirements, and memory needs.
- The TM4C123 series can still be used, but it would require a display module that includes its own display controller and driver. This approach may increase display cost and design complexity.
- TI's TivaWare Graphics Library provides functionality similar to basic graphics systems, including drawing primitives and touchscreen support. This may help reduce software development effort for simple display applications.
- A TM4C129 LaunchPad development board and/or compatible display BoosterPack has been ordered for testing.
- Mike or Toby will begin testing communication between the newer TM4C129 development platform and the Toro Diag system.

Conclusion

The current TM4C123-based display architecture is functional but limited by memory, graphics performance, and lack of an integrated LCD controller. As touchscreen display requirements increase, these limitations will make future development more difficult.

The TM4C129 platform appears to be the best first option to evaluate because it stays within the TI/Tiva ecosystem while adding integrated LCD controller support. However, memory limitations must still be reviewed, especially for higher-resolution displays.

The next major milestone is to prototype with the TM4C129 development hardware and verify Toro Diag compatibility. Once communication and display performance are tested, a more informed decision can be made on whether to continue with TM4C129, remain with TM4C123 and an external display controller, or migrate to a higher-performance microcontroller/display platform.